

Why Deep Neural Networks Work Well

CS4152 — Deep Learning and Neural Networks

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Outline

- 1 Hierarchical Feature Learning
- 2 Applications Beyond Images
- 3 Theoretical Justification
- 4 Practical Considerations
- 5 Conclusion

Compositional Representations

- Deep networks process data in hierarchical stages.
- Early layers: detect simple features (edges, wave patterns).
- Middle layers: form parts (eyes, noses, phonemes).
- Later layers: detect complex concepts (faces, words, sentences).
- Enables extraction of increasingly abstract representations.

Example: Face Recognition

- Layer 1: oriented edge detectors.
- Layer 2: edge combinations to detect facial parts.
- Layer 3+: assembling parts into full facial structures.
- Depth supports composition of simple $\rightarrow$ complex patterns.

- Input: raw audio waveforms.
- Early layers: tone, pitch, noise detection.
- Middle layers: phoneme detection (C–A–T).
- Later layers: word recognition → full phrases → sentences.
- Similar compositional hierarchy to vision tasks.

Circuit Complexity Intuition

- Some functions require exponential size if computed shallowly.
- Deep networks compute them with polynomial size.
- Example: XOR / parity of n bits.
- Deep XOR tree: depth $\mathcal{O}(\log n)$, small width.
- Shallow network: requires $\mathcal{O}(2^n)$ hidden units.

Shallow vs. Deep: Expressive Power

- Depth increases representational efficiency.
- Avoids brute-force enumeration of feature combinations.
- Enables learning functions with complex hierarchical structure.
- Explains success of modern deep models in many domains.

When to Use Deep Networks

- Not all tasks require dozens of layers.
- Good practice:
 - Start with logistic regression.
 - Try 1–2 hidden layers.
 - Increase depth as needed.
- Depth is a tunable hyperparameter.
- Excessive depth may lead to overfitting or training instability.

Depth vs. Branding

- “Deep Learning” partially succeeded due to powerful branding.
- But empirical performance validates the effectiveness of deep models.
- Depth enables hierarchical abstraction similar to biological systems.

Key Takeaways

- Deep networks learn hierarchical representations.
- Shallow networks may require exponentially more units.
- Hierarchical composition applies across modalities.
- Depth is powerful but must be tuned carefully.